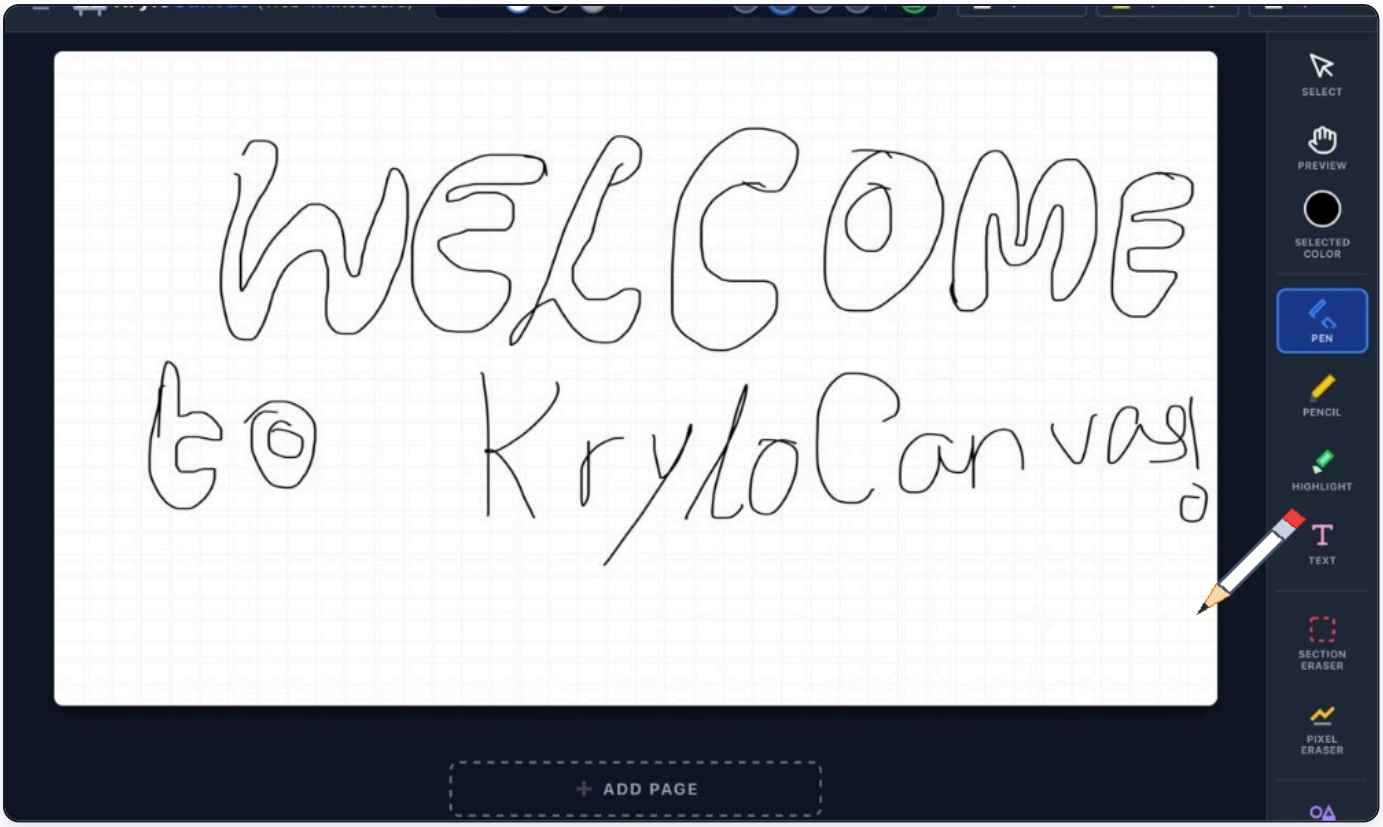
KryloCanvas

(Web-WhiteBoard) • Official Technical Guide

Software Architecture & Design by Abhimat Chatterjee



1. Executive Overview & Architecture

KryloCanvas (Web-WhiteBoard) is a professional, high-precision web-based digital whiteboard and mathematical drawing suite engineered for modern classrooms, STEM coaching institutes, and technical educators. Designed from the ground up to overcome the limitations of clunky desktop smartboard software, it runs natively inside any browser with zero backend latency, zero local installation, and full touch/stylus precision.

Core Operational Strengths:

- **Zero Dependencies:** Built on HTML5 Canvas, Fabric.js, PDF.js, and jsPDF.
- **Universal State Engine:** Real-time sync of color, thickness, and instrument geometry.
- **100dvh Adaptive Viewport:** Fluid scaling across iPads, touch monitors, and interactive flat panels.

Mathematical & STEM Capabilities:

- **Solid Boundary Snapping:** Magnetic pen guidance along rulers, protractors, and set squares.
- **Precision Geometry Compass:** Fixed needle pivot and infinite metric radius scaling.
- **43 Shapes & Process Trees:** 43 vector geometries and dynamic branching flowcharts.

2. Workspace Geography & Dual Operational Zones

KryloCanvas bifurcates its user interface into two specialized interaction zones designed for minimum cognitive load during live classroom teaching:

Top Command Header (System Bar)

- **File I/O Suite:** Import Images (PNG/JPG), multi-page PDF import via PDF.js worker, and vector PDF compilation via jsPDF.
- **Session History:** Dedicated 30-level undo/redo timeline tracked per canvas page.
- **Theme Inversion:** Instant switching across White, Black, and Studio Grey backgrounds with vector color flipping.
- **Paper Geometries:** Direct selection of Blank, 40px Grid, Ruled Notebook, and Dotted Matrix surfaces.
- **Engineering Modal:** Spawns 5 millimeter engineering graph paper palettes.
- **System Guards:** Modal-confirmed Refresh (🔄) and Close (✖) to prevent accidental loss.

Side & Bottom Floating Tool Palette

- **Pointer Selection:** Object transformation, group movement, scaling, and rotation via MTR handles.
- **Preview / Navigation:** Gesture-isolated two-finger pan and pinch zoom (50%–400%) without drawing.
- **Universal Swatch:** Live color indicator opening a 10-swatch palette, saturation slider, and HEX input.
- **Line Weight Calibrator:** Continuous thickness slider dynamically adjusting stroke width from 1px to 40px.
- **Drawing Instruments:** Pen, Pencil, Highlighter, Text, and Dual Erasers.
- **Technical Studios:** 43-Shape Studio, Solid Rulers, Compass, and Flowcharts.

3. Comprehensive Command Bar & Tool Switcher Reference

Interface Element	Trigger & Shortcut	System Behavior & Operational Impact
Select Mode	Sidebar / Key `V`	Enables bounding boxes on drawings, text, and flowchart blocks for editing or deletion.
Preview Mode	Sidebar `Preview` / Key `H`	Locks drawing surface; touch drag pans viewport across infinite canvas workspace.
Undo / Redo	Header / `Ctrl+Z` / `Ctrl+Y`	Reverts or restores previous vector stroke without disturbing active tools or background.
Add New Page	Bottom Bar `+ ADD PAGE`	Appends a clean canvas page to session array, incrementing dynamic header page counter.
Fullscreen Toggle	Header / Key `F11`	Expands canvas to 100% display area, hiding browser chrome on interactive flat panels.

4. Viewport Scaling, Dynamic Zoom & 100dvh Layout

100dvh Dynamic Scaling

Utilizes dynamic viewport units (`100dvh`) to adapt instantly when virtual on-screen keyboards, navigation bars, or browser chrome expand on touch screens.

Touch Gesture Isolation

Pinch-to-zoom and drag-pan are isolated to two-finger gestures or Preview tool, preventing accidental ink strokes while resting the palm on the display glass.

Pointer Event Decimation

Raw pointer input is filtered through an internal quadratic Bézier smoother at 60 FPS, eliminating hand tremors while preserving crisp, sharp corners.

Dynamic Canvas Resize Observers & Display Scaling

When running across classroom interactive flat panels (such as 65", 75", or 86" 4K Smartboards), the layout engine monitors window resize events using a debounced `ResizeObserver`. It automatically recalibrates CSS pixel density, re-centering the active drawing viewport and scaling the top command bar without re-rendering vector objects. This ensures educators never experience fuzzy lines or misaligned touch coordinate offsets when switching between full-screen presentation mode and standard windowed teaching views.

Multi-Page Session Management & Accidental Exit Protection

Educators can add unlimited sequential canvas sheets via the bottom `+ ADD PAGE` button. Each page maintains an independent Fabric.js canvas instance, complete history stack, and active tool bindings. System refresh (🔄) and close (✖) actions trigger custom safety dialogs to prevent accidental session termination.

5. The Universal Drawing & Styling Engine

KryloCanvas employs an atomic state synchronization engine (`appState`). Adjusting color or thickness dynamically propagates to active pens, pencils, inserted geometric shapes, and flowchart nodes.

Pen Tool (Natural Ink)

Delivers continuous, bold vector ink strokes matching the exact universal thickness and selected color. Optimized for stylus writing and rapid mathematical derivations.

Formula: `width = universalThickness`

Pencil Tool (Technical)

Simulates fine drafting graphite by automatically halving universal thickness (`max(1, thickness/2)`). Ideal for construction lines, geometry sketching, and diagrams.

Formula: `width = max(1, t/2)`

Highlighter (Translucent)

Dynamically converts solid hex colors into a 40% alpha wash (`rgba(r,g,b,0.4)`) at 4x thickness. Layered strokes never obscure underlying writing, text, or formulas.

Formula: `width = 4*t, alpha = 0.40`

6. Advanced Dual Eraser Suite: Marquee vs. Collision Eraser

Eraser Instrument	Selection Geometry	Deletion Mechanics	Ideal Educational Use Case
Section Eraser	Dashed rectangular marquee box dragged across canvas.	Atomic batch removal of all enclosed vector objects on mouse-up.	Clearing entire mathematical proofs, obsolete drawings, or large notes.
Pixel / Scribble Eraser	Continuous circular collision brush path.	Calculates line segment intersections and trims contacted strokes.	Surgically fixing incorrect exponents, minus signs, or small sketch lines.

7. Plain Text Annotation Tool (`IText`) Architecture

Vector Typography Engine

- **Tap-to-Type:** Instant instantiation of editable text nodes anywhere on canvas.
- **Live Color Sync:** Automatically inherits the active Universal Color upon creation.
- **Multi-Line Wrapping:** Supports standard newline breaks, carriage returns, and word wrapping.

Node Lifecycle & Auto-Pruning

- **Direct In-Place Editing:** Double-click existing text to alter strings; drag corner handles to scale.
- **Ghost Object Disposal:** Text boxes blurred without entered content are automatically pruned.
- **Vector Serialization:** Text preserves crisp font glyph rendering during vector PDF export.

8. Stroke Dynamics & Stylus Velocity Physics

Quadratic Bézier Smoothing & Tapering

The natural ink pen and pencil tools monitor pointer event pressure and movement velocity. When sketching rapid mathematical formulas, the Bézier engine generates intermediate control points: $Q(t) = (1-t)^2P0 + 2(1-t)tP1 + t^2P2$. The algorithm dynamically widens or narrows the vector path to simulate realistic friction and fountain pen taper, giving handwritten algebra, calculus integrals, and chemical structures a smooth, printed appearance.

Eraser Collision Mathematics & Segment Splitting

The scribble eraser treats the cursor path as a sequence of line segments with radius `R_eraser = universalThickness`. For every active path on the canvas, the collision engine calculates the segment-to-segment minimum distance: `dist(S1, S2) < R_eraser + strokeWidth/2`. When intersection is confirmed, the Fabric.js path object is split into discrete sub-paths or trimmed, guaranteeing immediate feedback without frame rate drops during vigorous scrubbing.

Typography Shortcuts & Multi-Line Alignment

While editing text, instructors can press `Enter` to create newlines, `Shift+Enter` for soft breaks, and use standard keyboard navigation (`Ctrl+A`, `Backspace`, `Delete`). Text automatically inherits system sans-serif font stacks for legibility across classroom projectors.

Color Swatch Engine & Line Weight Calibration

The universal color indicator opens a 10-swatch quick palette (Black, Red, Blue, Green, Yellow, Orange, Purple, Pink, Cyan, Grey) plus a full HSV color wheel and hex input field. Stroke thickness adjusts from 1px to 40px via a continuous slider, offering immediate visual feedback.

9. Indelible Header Stamp: Date, Day & Dynamic Page Counter

Every canvas page automatically renders an indelible header stamp at the top-right corner, ensuring complete temporal and pagination accountability for classroom exports:

Two-Line Header Stamp Layout

Top-Right Positioned Canvas Stamp:

```
13/10/2026 - Tuesday
4 / 10
```

- **Line 1 (Date & Weekday):** Formatted as `DD/MM/YYYY - DayName`. Automatically queries local system time, updating accurately across every day, month, and year.
- **Line 2 (Fractional Page Index):** Formatted in plain English digits as `X / Y` (e.g. `1 / 1`, `4 / 10`). The forward slash represents the integer division symbol requested for clear fractional tracking.
- **Dynamic Recalculation:** Inserting, removing, or reordering pages immediately invokes `updatePageHeaderStamp()`, recalculating total pages (`Y`) and updating every active canvas without reload.
- **Export Permanence:** The stamp is rendered directly into the canvas vector context, ensuring it remains permanently visible during image and PDF printing.

10. Theme Contrast Inversion Rules for Header Stamp

Active Theme	Background Color	Stamp Color Specification	Visual Contrast Rationale
White Theme	Clean White (`#ffffff`)	Translucent Black (`rgba(0,0,0,0.65)`)	Crisp daylight contrast without overpowering teacher annotations.
Black Theme	Dark Mode (`#000000`)	Translucent White (`rgba(255,255,255,0.72)`)	Gentle luminescent readability under dark classroom projection.
Grey Theme	Studio Grey (`#9ca3af`)	Opaque Solid White (`#ffffff`)	Maximum opacity to prevent wash-out against neutral grey tones.

11. Multi-Page Lifecycle & 30-Step History Stack

Per-Canvas Undo / Redo Serialization

Every canvas instance maintains an isolated `historyStack[]` and `historyIndex`. Each stroke completion serializes a JSON snapshot of the object tree. Undo operations restore the exact previous vector state without affecting other pages.

Dynamic Multi-Page Memory Management

Canvases not currently in view suspend intensive event listeners while keeping vector coordinates cached in memory. Adding 20+ pages maintains smooth 60 FPS performance without browser tab crashes or memory leaks.

12. Multi-Page Navigation & Session Management Protocols

Fast Sheet Switching & Object Caching

Navigating between pages in KryloCanvas is accomplished via the bottom navigation track and keyboard shortcuts (`← Page` and `Page →`). When transitioning between pages, the engine serializes the active canvas into a compact JSON state object and instantly deserializes the target canvas. This ensures zero latency during lecture delivery, allowing teachers to flip back and forth between lecture slides, homework problems, and student questions without losing any drawn vector paths.

Automatic Stroke Color Inversion Mechanics

When switching between White and Black themes, KryloCanvas iterates through all drawn objects. Pure black ink strokes (`#000000`) are converted to pure white (`#ffffff`), while colored strokes (red, blue, green, yellow) are retained with adjusted luminance to remain legible. Reverting back to White flips white strokes back to black, preserving full visual contrast without requiring manual redraws.

Indelible Vector Export Guarantee & Multi-Format Output

When the user triggers PDF Export, the engine iterates across all session pages in memory, renders each page onto an off-screen high-resolution vector canvas, burns in the top-right header stamp, and compiles a clean multi-page document via jsPDF. In addition, single-page snapshot export (`PNG`) produces crisp 2x retina images ideal for lecture slide decks and LMS assignment submissions.

When the user triggers PDF Export, the engine iterates across all session pages in memory, renders each page onto an off-screen high-resolution vector canvas, burns in the top-right header stamp, and compiles a clean multi-page document via jsPDF.

13. Mathematically Solid Instruments & Snapping Engine

Traditional digital whiteboards treat rulers as passive background images. KryloCanvas treats geometry instruments as **Mathematically Solid Physical Boundaries**. When drawing near an instrument edge, stroke coordinates snap to the nearest boundary edge, preventing ink bleedthrough.

Snapping Mathematics (`applySnappingLogic`)

The canvas pointer override detects active instruments within a threshold (`SNAP_THRESHOLD = 100px`). Every input point `P(x,y)` is projected orthogonally onto the nearest line segment or circular arc: `Proj = P1 + t * (P2 - P1)`, where `t = clamp(((P - P1) • (P2 - P1)) / |P2 - P1|², 0, 1)`. If distance `|P - Proj| < 100px`, the pointer locks to `Proj`, guaranteeing flawless straight edges and circular arcs.

The Solid Ruler

Dual Calibrated Scales: Top edge graduated in Metric (`cm/mm`); bottom edge in Imperial (`inches`). Calibrated at 96 DPI (`1 cm = 37.795 px`, `1 in = 96 px`).

Dynamic Extensibility: Drag side/corner handles to extend ruler length up to 100cm (3780px). Generates procedural vector tick marks and numbers without stretching text.

Solid 180° Protractor

Dual Degree Scales: Semi-transparent acrylic body with dual 0°–180° inner and outer angle rings and center crosshairs.

Polar Arc Snapping: Inking near the 250px semicircular rim locks stylus coordinates to radial angle: `x = cx + R*cos(θ)`, `y = cy - R*sin(θ)`.

Solid 45° Set Square

Orthogonal Precision: 45°–45°–90° triangle with hollow inner cutout and dual-leg metric rules.

Strict Alignment: Tracing horizontal, vertical, or hypotenuse edges yields perpendicular lines and exact 45° angles for engineering drawings.

14. Instrument Management & Custom Controls

Control Action	Interaction Method	Operational Behavior
Translate / Move	Switch to `Select`, click body, and drag.	Moves entire instrument without affecting underlying drawings or text.
Rotate (`MTR`)	Drag blue circular handle at top center.	Rotates instrument around its origin; degree snapping reflects active angle.
Scale / Elongate	Drag blue corner or side bounding boxes.	Rulers re-render procedural vector markings to match target width.
Delete Control	Click red trash icon (✖) at top-right corner.	Instantly dismisses instrument; underlying drawings remain untouched.
Export Protection	Automated via `isUIElement: true` tag.	Instruments auto-hide during PDF export and restore immediately afterward.

Procedural Millimeter Tick Mark Mathematics

When extending the Solid Ruler up to 100 cm, vector tick marks are procedural, not stretched. The tick rendering loop calculates tick positions via: `x = origin + i * (PX_PER_CM / 10)`. Major centimeter ticks (10 mm) render at 18px height with bold numeric labels; half-centimeter ticks (5 mm) render at 12px height; and individual millimeter ticks (1 mm) render at 7px height. This procedural generation guarantees absolute sharpness regardless of ruler length.

Scale Invariance During Viewport Pan & Zoom

Physical instrument calibration is locked to Euclidean screen coordinates. Panning or zooming the board does not deform instrument measurements: 1 cm on the ruler always corresponds to 37.795 screen pixels. Multi-instrument placement allows teachers to position the Solid Ruler, 180° Protractor, and Set Square simultaneously on the same board without layout interference.

Pedagogical Pro-Tip: Teaching Geometry with Magnetism

To demonstrate geometric constructions (e.g. bisecting angles, constructing tangents, or drawing coordinate axes), spawn the Solid Ruler and Protractor side-by-side. Use the Pen tool to trace boundary lines; the cursor will snap to ruler edges and arc perimeters without ruler bleedthrough, producing presentation-grade board work.

15. Precision Geometry Compass System

Located directly beneath the Rulers tool in the side toolbar, the **Precision Geometry Compass** provides a complete physical and mathematical simulation of a technical drafting pair of compasses.

Compass Anatomy & Construction

- **Top Hinge & Grip:** Knurled cylinder handle and brass joint connecting both compass arms.
- **Fixed Needle Leg (Left):** Tapered stainless steel beam terminating in a sharp needle point and red bullseye crosshair (`NEEDLE (CENTER)`).
- **Pencil Leg (Right):** Mechanical clamp holding a wooden pencil with a graphite lead tip reflecting the active Universal Color.
- **Connecting Dimension Arc:** Dashed guide arc and dimension line displaying calibrated radius in real-time.

Mathematical Scale Calibration

- **Pixel-Metric Lock:** Strictly calibrated to `PX_PER_CM = 37.795` and `PX_PER_INCH = 96`, matching the Solid Ruler with zero measurement error.
- **Infinitely Extensible Arm:** Unlike physical school compasses with limited reach, the virtual arm extends from 0.5 cm to 500+ cm across any canvas.
- **Fixed Needle Preservation:** Extending or rotating the arm recalculates matrix offsets so the center coordinate never drifts: `ΔPos = P_needle - Transform(Local_Needle)`.

16. Coordinate Invariance & Needle Anchor Math

Mathematical Proof of Center Point Invariance

In standard interactive whiteboards, rotating a compound object causes its visual needle point to revolve around the group's geometric bounding centroid, displacing the center of the drawn circle. KryloCanvas eliminates this flaw through dynamic affine matrix inversion. Let `P_needle` be the user's fixed center coordinate in world space. When the pencil arm rotates by angle `θ` or expands by radius `ΔR`, the engine computes the required group translation: `T_new = P_needle - R(θ) × LocalNeedleOffset`. The group's top-left origin is repositioned by `T_new`, locking the needle point to the exact same screen pixel with zero drift (< 0.01 px).

17. Mechanical Dimensions & Calibration Specifications

Compass Component	Material / Visual Styling	Scale Ratio / Metric Range	Operational Function
Knurled Top Grip	Knurled brass knurling (12px × 36px)	Fixed aspect ratio anchor	Provides primary touch drag handle for whole instrument translation.
Needle Leg (Fixed)	Tapered stainless steel (140px length)	Locked pivot point	Red bullseye crosshair pinned permanently to user center coordinate.
Pencil Leg (Lead)	Cedar casing with graphite tip	Dynamic universal color	Renders swept vector arcs and circle circumferences matching active pen color.
Dimension Guide Arc	Dashed circular guide line	1 cm = 37.795 pixels	Displays real-time Euclidean radius measurement label in cm and mm.
Extensible Arm	Telescoping horizontal guide	0.5 cm to 500+ cm range	Expands arm infinitely across single or multi-page canvases without boundary clipping.

Knurled Grip Physics & Whole-Assembly Translation

The knurled top cylinder acts as an omnidirectional gimbal handle. Clicking or touching the top handle translates the entire compass assembly across the screen, keeping the current radius opening and needle-pencil angle rigid. This allows teachers to pick up the compass from one geometric construction and place it onto another without having to re-measure.

Responsive Mobile & Tablet Ergonomics & Keyboard Shortcuts

Controls are calibrated for finger tap targets on iPads, Android tablets, and large interactive screens. Desktop users can also control the compass via keyboard: press `[` and `]` to decrease or increase radius by 0.5 cm, and `<` or `>` to rotate the arm in 15° increments. Controls are calibrated for finger tap targets on iPads, Android tablets, and large interactive screens. Sliders feature smooth inertia damping to prevent accidental over-extension during demonstrations. Spawning the compass automatically opens a quick instruction dialog that can be dismissed with a single tap.

18. Interactive Compass HUD & Operational Controls

Spawning the compass activates the floating **Compass HUD**, providing precision digital controls:

HUD Control	Action & Input Format	Operational Function
Radius Display	Live readout: `R = 5.00 cm (50.0 mm)`	Displays exact Euclidean distance between needle pivot and pencil lead.
Steppers & Slider	Buttons `-0.5cm` / `+0.5cm`, slider (1–40cm)	Adjusts radius smoothly in 0.5cm steps or continuous sub-millimeter fractions.
Numeric Input	Direct text input box: `[ 5.0 ] cm`	Type any arbitrary radius (e.g. 12.75 cm); arm recalculates and locks instantly.
Arm Rotation	Buttons `↺ 15°` / `↻ 15°`	Rotates pencil arm around the fixed needle point in 15° calibrated increments.
Draw Full Circle	Button `⭕ Circle`	Generates a mathematically perfect 360° vector circle (`fabric.Circle`) at current radius.
Draw Swept Arc	Button `📐 Arc (90°)`	Stamps an exact 90° circular arc path starting from current pencil position.
Stylus Snapping	Draw near circumference with Pen or Pencil.	Pen point locks magnetically to circumference: `P = C + R * unit(P - C)`.

19. Classical Geometric Construction Blueprints

Perpendicular Bisector Workflow

1. Place needle on segment endpoint A; set radius greater than half length (e.g. 6.0 cm).
2. Click `📐 Arc` to stamp a 90° arc above and below segment.
3. Move needle to endpoint B with identical radius; stamp intersecting arcs.
4. Use Solid Ruler to connect arc intersection points for exact 90° bisection.

Angle Bisector Construction

1. Place needle on angle vertex V; stamp an arc cutting both angle rays at points X and Y.
2. Place needle on X and Y successively to stamp intersecting arcs in the angle interior.
3. Draw ray from V through the intersection using the ruler for an exact 50% angle split.

Equilateral Triangle Construction

1. Draw base segment AB using the Solid Ruler. Place compass needle on A, set radius to AB.
2. Stamp an arc above the segment. Move needle to B and stamp an intersecting arc at C.
3. Connect C to A and B with the Solid Ruler for an exact 60° equilateral triangle.

Circle Tangent from External Point P

1. Connect external point P to circle center O; bisect segment OP to find midpoint M.
2. Place needle at M with radius MO; stamp arcs cutting the circle at T1 and T2.
3. Draw rays PT1 and PT2 using the Solid Ruler to construct the two exact tangents.

Construction 5: Inscribing a Regular Hexagon in a Circle

1. Draw target circle with radius R. Pick starting point P0 on circumference.
2. Place needle on P0 with radius R; stamp an arc cutting circumference at P1.
3. Step needle to P1, P2, P3, P4, P5 consecutively, stamping 6 equidistant vertices.
4. Connect adjacent vertices with Solid Ruler for a mathematically perfect regular hexagon.

Stylus Circumference Snapping Mechanics

In addition to 1-click button stamping, teachers can pick up the Pen or Pencil tool and sketch near the compass circumference. The pointer projection algorithm locks the stylus tip onto the circle perimeter (`P_lock = C + R × unit(P - C)`), allowing teachers to manually sweep custom arc lengths without deviating from the true circle radius.

20. The Extended 43-Shape Library

Accessible via the Shapes button in the sidebar, KryloCanvas features **43 procedural geometric figures**, directional arrows, technical drafting lines, and flowchart symbols.

Category	Included Shapes	Primary Educational & Technical Applications
Triangles (4)	Equilateral, Isosceles, Scalene, Right-Angled Triangle.	Pythagorean theorem demonstrations, trigonometry, angle sums, and congruence proofs.
Quadrilaterals (6)	Square, Rectangle, Parallelogram, Rhombus, Trapezium, Kite.	Area and perimeter proofs, parallel lines, quadrilateral properties, and shearing models.
Regular Polygons (4)	Pentagon (5), Hexagon (6), Heptagon (7), Octagon (8).	Interior angle calculations: `(n-2)*180°/n`, symmetry axes, and tessellations.
Curved Solids (6)	Circle, Semicircle, Oval, Rounded Rect, Rounded Square, Squircle.	Circular geometry, ellipse focal points, superellipse contours, and rounded UI wireframing.
Technical Lines (3)	Straight Line (160px), Curved Bezier Line (arc), S-Curve Wave.	Transversal lines, coordinate axes, polynomial curves, wave mechanics, and inflection points.
Standard Arrows (4)	Arrow Right (→), Arrow Up (↑), Arrow Left (←), Arrow Down (↓).	Vector force diagrams, physics ray tracing, chemical reaction progressions, and dimension markers.
Complex Arrows (4)	Double Arrow H (↔), Double Arrow V (⇵), Curved Arrow R, Curved Arrow L, Return Arrow.	Equilibrium reactions, feedback loops, rotational torque, reversible thermodynamic cycles, and U-turns.
Flowchart Symbols (4)	Diamond (Decision), Cylinder (Database), Document (Print), Data / IO (Parallelogram).	Standard ISO process mapping, algorithm flowcharts, data input/output, and database architectures.
Creative Symbols (8)	Star (5-point), Cross, Heart, Moon (crescent), Cloud, Callout (Speech Bubble), Wedge.	Classroom annotations, dialogues, astronomy diagrams, grading marks, and thought bubbles.

21. Procedural Geometry Construction Formulas

Shape Designation	Mathematical Formulation / Parametric Path	Key Geometric Properties
Squircle (Superellipse)	Lamé Curve: `|x/a|&sup4; + |y/b|&sup4; = 1` (`rx: 34, ry: 34`)	Continuous curvature with zero abrupt corner transitions.
Quadratic Bézier Arc	Parametric: `P(t) = (1-t)²P0 + 2(1-t)tP1 + t²P2`	Dynamic curvature for projectile motion and optics.
Regular Hexagon	Polar Vertices: `x_k = cx + r*cos(2πk/6)`	Exact 120° interior angles; ideal for aromatic chemistry rings.
Decision Diamond	Rotated Rhombus: `M 0 h/2 L w/2 0 L w h/2 L w/2 h Z`	ISO algorithm decision branching symbol.

22. Shape Transformation & Aspect Ratio Locking

Bounding Handle Manipulation & Rotation Snapping

When any shape is selected with the Select tool, 8 corner and edge handles appear. Dragging corner handles scales the shape uniformly when holding Shift, while dragging edge midpoints stretches width or height independently. The top blue rotation handle (MTR) rotates the shape with 15° angle snapping when holding Shift.

Cross-Disciplinary STEM Classroom Applications

Physics & Mechanics: Combine Straight Lines and Standard/Double Directional Arrows to construct free-body force diagrams, gravitational vectors, and normal reaction forces with exact angular alignment. **Organic Chemistry:** Utilize regular Hexagon and Pentagon shapes to draw aromatic rings, cyclohexane conformations, and complex molecular skeletons with uniform bond lengths. **Computer Science:** Assemble decision diamonds, I/O parallelograms, and database cylinders to illustrate algorithms before code execution.

22. Flowchart & Process Tree Builder Engine

Initiated via the `Flowchart` button in the sidebar, this engine empowers educators to build structured algorithmic pipelines, organizational charts, and hierarchical concept trees without manual box alignment.

1. Flow Orientation

- **Horizontal Flow (`→`):** Left-to-right sequential process pipelines; ideal for timelines, workflows, and chemical equations.
- **Vertical Flow (`⇵`):** Top-to-bottom hierarchy trees; ideal for classification trees, family trees, and logic branches.

2. Structure Templates

- **Linear Process:** 2 to 8 linked sequential blocks with auto-aligned connecting arrows.
- **Tree Hierarchy:** 1 Root Node → 2 Subordinate Branches → 4 Leaf Nodes.
- **Single Node:** 1 standalone root block ready for freeform expansion.

3. 6 Block Color Themes

- **Ocean Blue:** Standard process steps.
- **Emerald Green:** Success / start nodes.
- **Purple Tech:** Computing & data blocks.
- **Sunset Orange:** Action / warning blocks.
- **Slate Minimal:** B&W LaserJet friendly.
- **Universal:** Active drawing color.

23. Interactive In-Page Editing & Floating Flowchart HUD

Selecting any flowchart node summons the floating **Flowchart HUD** (`#flowchart-hud`), providing high-speed node management:

HUD Action	Trigger Mechanism	Engine Execution & Visual Outcome
Add Connected Step	Button `+ Add Step`	Spawns next sequential block along flow axis; draws an arrow from parent block.
Branch Leaf Node	Button `🌿 Leaf Node`	Spawns a subordinate child node offset diagonally with a connecting directional branch arrow.
Edit Block Text	Button `abc Text` or **Double-Click**	Opens in-page modal dialog (`#modal-fc-text`) to update block label with zero page reload.
Delete Node	Button `🗑 Delete`	Removes selected block from canvas; cleans up incoming and outgoing connector lines.
Toggle Alignment	Button `↔ / ⇵ Dir`	Switches growth orientation between horizontal and vertical axes for future appended nodes.
Quick Palette	5 Color Swatch Dots on HUD	Instantly re-themes selected block or whole branch (Blue, Green, Purple, Orange, Slate).

24. High-Contrast Block Color Themes Specifications

Theme Name	Background Fill	Outline Border	Primary Educational & System Application
Ocean Blue	Sky Light (`#e0f2fe`)	Deep Sky (`#0284c7`)	Standard operational stages, data flow, and sequential algorithm steps.
Emerald Green	Mint Light (`#dcfce7`)	Forest (`#16a34a`)	Process initiation, entry states, successful validation, and base cases.
Purple Tech	Lavender (`#f3e8ff`)	Deep Purple (`#9333ea`)	Mathematical functions, subroutines, recursion, and database queries.
Sunset Orange	Peach Light (`#ffedd5`)	Flame (`#ea580c`)	Decision outcomes, conditional warnings, exceptions, and loops.
Slate Minimal	Off-White (`#f8fafc`)	Black (`#0f172a`)	Black & white laser printing and clean document-friendly handoffs.

Gesture Physics: Pull-to-Extend & Push-to-Shrink

In addition to button clicks, the flowchart engine implements continuous bounding box gesture tracking on `object:modified`: dragging any handle outward (`scaleX > 1.3` or `scaleY > 1.3`) resets scale to 1.0 and **automatically appends a new connected subordinate node** in that direction. Pushing handles inward (`scale < 0.75`) safely removes the node.

25. Canvas Themes, Page Styles & Graph Papers

KryloCanvas supports 3 contrast themes, 4 standard page paper styles, and 5 millimeter engineering graph colors:

Themes & Contrast Inversion

- **White Theme:** Clean daylight background (`#ffffff`).
- **Black Theme:** Dark mode (`#000000`); automatically scans and inverts black strokes to white.
- **Grey Theme:** Neutral low-glare studio background (`#9ca3af`).
- **Stroke Inversion:** Reverting back to White flips white strokes back to black automatically.

Paper Styles & 5 Graph Papers

- **Blank Page:** Pure unlined paper for art and freeform notes.
- **Standard Grid:** 40px square grid for alignment.
- **Single Lines:** 36px notebook ruled lines for handwriting.
- **Dotted Matrix:** 36px bullet journal dot grid.
- **5 Graph Palettes:** Classic Green, Blueprint Blue, Red/Maroon, Amber Orange, and High-Contrast B&W.

26. Official System Access & Software Credentials

KryloCanvas is freely accessible worldwide as a lightweight web application. To launch the full whiteboard studio:

Direct Web Deployment

krylocanvas.vercel.app

Point your smartphone, tablet, or interactive flat panel camera at the QR code on the left to immediately load the studio. Compatible with Chrome, Safari, Edge, Firefox, Android, and iOS.

Website URL: https://krylocanvas.vercel.app

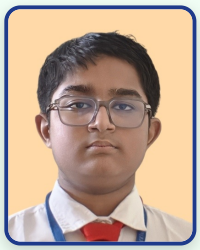
Software Architecture & System Engineering

About the Creator: Abhimat Chatterjee is an independent developer interested across multiple fields including **Robotics, STEM disciplines, algorithms, and interactive educational software**.

System Development: Driven to overcome the clunky latency and lack of precision in traditional digital boards, it was conceptualized, designed, and coded from scratch as a lightweight, zero-dependency web application. It features a high-performance vector rendering pipeline, mathematical boundary-projection snapping for physical instruments, an extensible metric compass, dynamic flowchart trees, and 43 geometric drafting shapes.

© Copyright 2026 Abhimat Chatterjee. All rights reserved.
This instruction manual, system specification, and the KryloCanvas software engine are the proprietary intellectual property of the creator. No portion of this code, interface architecture, mathematical snapping algorithms, or documentation may be copied, distributed, sold, or replicated without explicit written permission from the creator.

Abhimat Chatterjee
LEAD ARCHITECT & CREATOR



Photograph of the Creator

KryloCanvas (Web-WhiteBoard) • Built to empower teachers, inspire students, and elevate classroom instruction through mathematical rigor and digital simplicity.